ANKUSH MANHAS

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Junior Machine Learning Engineer | Python | Time Series | Leakage Checks | scikit-learn

PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech, 2026) who writes Python every day and builds machine learning projects end to end. Currently building a fraud detection system on 24 million time-ordered card transactions, with a time-based train and validation split to keep the future out of the training set, and a test on 253 people the model never saw. Doubts a good result before trusting it: dropped a dataset after finding columns that leak the label, and removed a strong feature after showing it drifted over time.

EDUCATION

Chandigarh University — May 2026
B.Tech, Computer Science — CGPA: 7.2 / 10.0

SKILLS

Python: pandas, NumPy, scikit-learn, XGBoost, Matplotlib, Jupyter, Streamlit, modular code (src/ packages)
Machine Learning: train and test split, time-based validation for time series, cross-validation, overfitting diagnosis, leakage detection, feature engineering, logistic regression, random forests, class imbalance, PR-AUC, precision and recall, error analysis, drift analysis
Data: data cleaning, alignment in time, data-quality checks, chunked loading of files larger than memory, Parquet, SQL (joins, window functions)
Also: Git, GitHub, Linux, Java (data structures and algorithms)

PROJECTS

Real-Time Fraud Detection System — Ongoing | Python, pandas, scikit-learn, Parquet

- Working on IBM TabFormer card transactions (24.4 million rows, about 12 frauds in 10,000). github.com/Ankush-Manhas840/fraud-detection-realtime
- Found leakage in the first dataset (PaySim): balance columns were never updated for fraud rows, so they gave away the label; also senders never repeat, so per-person history was impossible. Switched datasets.
- Split train and validation by time (cutoff April 2017; 2.4 million training rows) instead of shuffling, and built per-card history features (minutes since last purchase, new shop, hour, outside the United States, amount).
- Logistic regression baseline: PR-AUC 0.090, 150 real frauds in its top 1,000 alerts against about 1 for random. Reproduced on 253 unseen people in a later period: PR-AUC 0.074, 142 in the top 1,000.
- Showed that adding chip/online card use made the model worse: online fraud fell from 66 to near 0 per 10,000 after 2017, confirmed in a second independent group of people, so the feature was left out.
- Moved notebook logic into reusable src/data.py and src/features.py (checked equal to notebook output). Next: XGBoost, experiment tracking, a FastAPI scoring service.

Crop Yield Prediction — Deployed | Python, scikit-learn, pandas, Streamlit

- Predicts yield for 4 Indian crops from 246,091 rows of government data; test MAE fell from 107.63 (linear baseline) to 2.04 t/ha. github.com/Ankush-Manhas840/crop-yield-prediction | live demo
- Found crops recorded in non-tonne units with no column saying so (a 175x jump between median and third quartile) by checking quantiles, and narrowed scope to four crops with consistent units.
- Broke error down per crop (one crop gave 65.15 MAE against 0.55 for the rest), traced it to impossible outlier rows, and cut overall error by 83%.

Used Car Price Prediction — Deployed | Python, scikit-learn, pandas

- Random forest on about 15,400 CarDekho listings: test R^2 0.94 (linear baseline: 0.79) and test MAE ₹0.92 lakh; 5-fold cross-validation R^2 0.90–0.95. github.com/Ankush-Manhas840/used-car-price-predictor | usedcarchecker.streamlit.app
- Ranked the largest errors and found one ₹3.95 crore supercar dominating the error; removed brands with fewer than 10 listings (9 of 15,407 cars), raising R^2 from 0.72 to 0.94.

ONGOING WORK

- Daily data structures and algorithms practice in Python: github.com/Ankush-Manhas840/Practice | System design notes: github.com/Ankush-Manhas840/system-design